Chapter 4 • Factorization and Algebraic Manipulation

Solved Exercise 4.4

1. Find square roots by factorization.

(i) $x^2-8x+16=(x-4)^2$, so the square roots are $\pm(x-4)$.
(ii) $9x^2+12x+4=(3x+2)^2$, so the square roots are $\pm(3x+2)$.
(iii) $36a^2+84a+49=(6a+7)^2$, so the square roots are $\pm(6a+7)$.
(iv) $64y^2-32y+4 = 4(16y^2-8y+1)$
$= 4(4y-1)^2$
$= [2(4y-1)]^2$
$= (8y-2)^2$.
$\pm(8y-2)$
(v) $200t^2-120t+18 = 2(100t^2-60t+9)$
$= 2(10t-3)^2$.
$\pm\sqrt{2}(10t-3)$
(vi) $40x^2+120x+90 = 10(4x^2+12x+9)$
$= 10(2x+3)^2$.
$\pm\sqrt{10}(2x+3)$.

2. Find square roots by the division method.

(i) Division gives $2x^2-7x-3$; check: $(2x^2-7x-3)^2=4x^4-28x^3+37x^2+42x+9$.
$\pm(2x^2-7x-3)$
(ii) $(11x^2-9x-12)^2=121x^4-198x^3-183x^2+216x+144$.
$\pm(11x^2-9x-12)$
(iii) $(x^2-5xy+y^2)^2=x^4-10x^3y+27x^2y^2-10xy^3+y^4$.
$\pm(x^2-5xy+y^2)$
(iv) $(2x^2-3x+7)^2=4x^4-12x^3+37x^2-42x+49$.
$\pm(2x^2-3x+7)$.

3. $R(x)=-x^2+6x-8$. Find investment levels giving zero return.

$R(x) = -(x^2-6x+8)$
$= -(x-2)(x-4)$.
$R(x)=0$ when $x-2=0$ or $x-4=0$.
$x=2$ or $x=4$ thousand rupees.

4. $P(x)=x^3-15x^2+75x-125$. Find break-even points.

$P(x)=(x-5)^3$ because $(x-5)^3=x^3-15x^2+75x-125$.
$x=5$ units, a repeated break-even root.

5. $V(x)=2x^3-6x^2+4x$. Find where $V(x)=0$.

$V(x) = 2x(x^2-3x+2)$
$= 2x(x-1)(x-2)$.
$x=0$, $x=1$ or $x=2$.

6. $Y(x)=2x^2-8x+6$. Find points of zero deflection.

$Y(x) = 2(x^2-4x+3)$
$= 2(x-1)(x-3)$.
$x=1$ or $x=3$.

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